Partitioning the first conjugacy classes

Human Compression project

We acquire the reviewed prefix enumeration and its exact implementation. For a fixed literal permutation group G, let $(a_1, \ldots, a_s)$ be the complete ordered list of conjugacy class representatives of order greater than two used by that implementation. Its outermost loop starts the prefix $(1, a_i)$ once for each i. All later searches use the centralizer of the current prefix and the unchanged intersection tests.

Proposition 1. *Executing those s outermost branches separately and taking the disjoint union of their returned lists gives the same complete collection of inner orbits as the acquired enumeration. This remains true when a completed branch is acquired from an earlier execution, provided its exact group, ordered representative list and implementation are bound.*

Proof. The recursive visit uses only the fixed group, its involutions, the current prefix and its parabolic subgroups. Earlier results and counters are never read in selecting an extension or testing a marking. Restricting the outer loop to one index therefore leaves its complete mathematical output unchanged. An inner automorphism relating two resulting markings must conjugate their first nonidentity entries. Distinct entries of the list represent distinct conjugacy classes, so different branches cannot represent the same inner orbit. Conversely every admissible generating marking has a first nonidentity entry in one of those classes. The acquired completeness theorem places its orbit in that branch. □

Each returned file includes the whole ordered representative list, the chosen index and the exact literal marking arrays. The collector requires identical literal group inputs and representative lists, identical centre and automorphism orders, and exactly the indices $1, \ldots, s$, each once. It checks that every returned marking starts at its branch representative. If a group algorithm chooses different representatives on another run, the collector rejects that join; an index alone is never its identity. No previous interrupted branch supplies a count. A deadline leaves that branch unpaid and does not erase completed files for other branches.

After the complete union, the acquired free-action argument divides the inner count in each rank, ordered type and chirality class by $|\operatorname{Out}(G)|$. This division is not made within an individual first branch: an outer automorphism can exchange first conjugacy classes. The regular entries in this output still mean directly regular rotation groups. They are not the regular columns of the almost-simple source atlas.

The recursion, group arithmetic and full-automorphism computation retain their earlier proofs and software premises. The present change is only the finite partition and exact receipt union. The writer uses a wider GAP output line to avoid adding line-continuation escapes to JSON fields. No target count, source admission, new conjecture or formalization follows from the method alone.